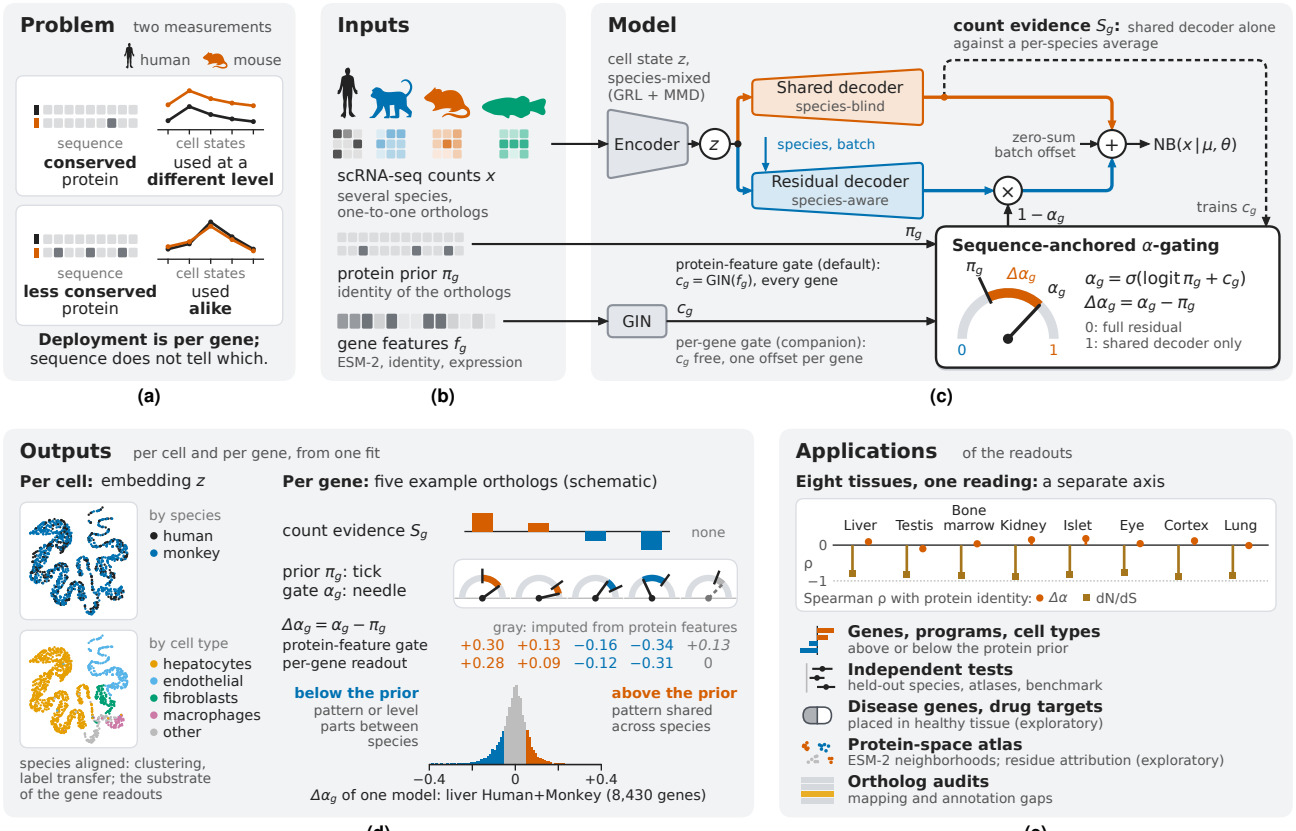

Problem
two measurements
human
mouse
sequence
conserved protein
cell states
used at a different level
sequence
less conserved protein
cell states
used alike
Deployment is per gene; sequence does not tell which.
(a)
Inputs
scRNA-seq counts x
several species, one-to-one orthologs
protein prior π_g
identity of the orthologs
gene features f_g
ESM-2, identity, expression
(b)
Model
cell state z, species-mixed (GRL + MMD)
Encoder
z
Shared decoder
species-blind
species, batch
Residual decoder
species-aware
count evidence S_g: shared decoder alone against a per-species average
zero-sum batch offset
NB(x | μ, θ)
1 − α_g
trains c_g
π_g
Sequence-anchored α-gating
protein-feature gate (default):
c_g = GIN(f_g), every gene
GIN
c_g
per-gene gate (companion):
c_g free, one offset per gene
π_g
Δα_g
α_g
0
1
α_g = σ(logit π_g + c_g)
Δα_g = α_g − π_g
0: full residual
1: shared decoder only
(c)
Outputs
per cell and per gene, from one fit
Per cell: embedding z
by species
human
monkey
by cell type
hepatocytes
endothelial
fibroblasts
macrophages
other
species aligned: clustering, label transfer; the substrate of the gene readouts
Per gene: five example orthologs (schematic)
count evidence S_g
none
prior π_g: tick
gate α_g: needle
Δα_g = α_g − π_g
gray: imputed from protein features
protein-feature gate +0.30 +0.13 −0.16 −0.34 +0.13
per-gene readout +0.28 +0.09 −0.12 −0.31 0
below the prior
pattern or level parts between species
above the prior
pattern shared across species
−0.4
0
+0.4
Δα_g of one model: liver Human+Monkey (8,430 genes)
(d)
Applications
of the readouts
Eight tissues, one reading: a separate axis
Liver
Testis
Bone marrow
Kidney
Islet
Eye
Cortex
Lung
0
ρ
−1
Spearman ρ with protein identity: Δα dN/dS
Genes, programs, cell types
above or below the protein prior
Independent tests
held-out species, atlases, benchmark
Disease genes, drug targets
placed in healthy tissue (exploratory)
Protein-space atlas
ESM-2 neighborhoods; residue attribution (exploratory)
Ortholog audits
mapping and annotation gaps
(e)